BUSINESS ANALYSIS AND INSIGHTS

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Data Explanation

This analysis was generated from 5 dataset which consist of:

homepass_fin		
Column Name	**Data Type**	**Notes**
homeid	Object	The unique ID for every house passed by the fiber network
region	Object	The geographical location (e.g., Jakarta Utara, Bekasi). Used for location-based performance analysis
fibernode	Object	The specific network node or distribution point where the fiber optic cable originates for a certain area
technology	Object	The specific network node or distribution point where the fiber optic cable originates for a certain area
build_date	Object	The date when the network infrastructure in that specific node/area was completed
capex_cost	Int	Capital Expenditure; the initial investment cost to build the infrastructure in that area.
exclusive_flag	Object	A marker ('Y'/'N') indicating if your company is the sole provider in that area
exclusive_servco_id	Float	The ID of the Servco that has the sole right to sell services in that specific area
exclusive_start_date	Object	The timeframe for which the exclusivity valid agreement start
exclusive_end_date	Object	The timeframe for which the exclusivity valid agreement end

media_package_fin		
Column Name	**Data Type**	**Notes**
contract_account	Object	Used to link media purchases back to the main internet subscription
catv_package	Object	The specific Cable TV or Media package assigned to the customer's account
add_on_product	Object	The name of the specific media service
package_price	Int	The base price of the media plan

media_package_fin		
Column Name	**Data Type**	**Notes**
ao_wholesale_price	Int	The "cost of goods sold" (COGS) or the price your company pays to the provider
ao_rrp_price	Int	The Recommended Retail Price (RRP) for the specific add-on product

region_master_fin		
Column Name	**Data Type**	**Notes**
region	Object	The broad geographical area
island	Object	The island where the infrastructure is located
urban_rural	Object	A classification of whether the area is a densely populated city (Urban) or a countryside/remote area (Rural).
fibernode	Object	The specific network node or distribution point where the fiber optic cable originates for a certain area

servco_fin		
Column Name	**Data Type**	**Notes**
servco_id	Int	A unique identifier for each Service Company (Servco)
servco_name	Object	The display name of the partner
contract_start	Object	
minimum_guarantee	Float	The minimum revenue or number of subscribers that a Servco must guarantee to the infrastructure owner for that specific area.
lease_fee_per_active	Int	The fixed revenue the company earns for every active subscriber managed by that Servco

subscription_snapshot_fin		
Column Name	**Data Type**	**Notes**
snapshot_date	Object	The specific date/timestamp when the data was captured (usually end-of-month)
homeid	Object	The unique identifier for the physical house/location (Homepass)

subscription_snapshot_fin		
Column Name	**Data Type**	**Notes**
servco_id	Int	A unique identifier for each Service Company (Servco)
contract_account	Object	The unique ID for each customer's contract
status	Object	The current state of the subscription
activation_date	Object	The date the service actually went "live" and the customer started being charged
disconnect_date	Object	The date the customer stopped the service. Used to calculate Churn Rate.
active_flag	Int	Indicator to show if the account is currently generating revenue
bandwith_mbps	Int	The maximum internet speed capacity (in Megabits per second) allocated to a specific contract or node

Section 1 – Infrastructure & Utilization

1. Calculate montly penetration rate (Active CA/Total Homepass) by Region, Technology, and Exclusive vs Non-Exclusive Areas

To supplement the analysis of the penetration data:

- FTTH (Fiber To The Home): Employs end-to-end fiber optic cables directly to the premises, providing the highest possible bandwidth and lowest latency.
- HFC (Hybrid Fiber Coaxial): A legacy-friendly technology that combines fiber optics with coaxial cables. It is often the 'first mover' in many residential areas due to its history with cable TV networks.

	snapshot_date character varying (200)	region character varying (200)	technology character varying (200)	area_type text	active_ca bigint	total_hp bigint	penetration_rate_pct numeric
1	2025-12-31 0:00:00	Bekasi	HFC	Non-Exclusive	28	29	96.55
2	2025-12-31 0:00:00	Tangerang	HFC	Non-Exclusive	27	29	93.10
3	2025-12-31 0:00:00	Jakarta Utara	HFC	Non-Exclusive	21	23	91.30
4	2025-12-31 0:00:00	Jakarta Utara	FTTH	Non-Exclusive	68	77	88.31
5	2025-12-31 0:00:00	Jakarta Selatan	HFC	Non-Exclusive	28	32	87.50
6	2025-12-31 0:00:00	Bekasi	FTTH	Non-Exclusive	62	71	87.32
7	2025-12-31 0:00:00	Depok	FTTH	Non-Exclusive	63	73	86.30
8	2025-12-31 0:00:00	Jakarta Timur	HFC	Non-Exclusive	25	29	86.21
9	2025-12-31 0:00:00	Jakarta Utara	HFC	Exclusive	11	13	84.62
10	2025-12-31 0:00:00	Jakarta Pusat	FTTH	Non-Exclusive	26	31	83.87

The highest penetration rate is found in Bekasi (Non-Exclusive) using HFC technology at 96.55%. This indicates that in high-density suburbs, reliable broadband for basic daily needs remains the primary driver. The dominance of Non-Exclusive areas in the top 10 suggests that market competition is effectively pushing penetration rates higher than in Exclusive zones.

2. Identify High Capex & Low Utilization areas AND Underperforming Fibernodes

To understand our finding better, here are some information related to the question:

- High CAPEX: Refers to fibernodes with an upfront investment cost that exceeds the company's average capital expenditure for infrastructure deployment.
- Utilization Rate (Util_pct): The percentage of active subscriber accounts relative to the total homepasses available.
- Network Average Utilization (Network_avg_util_pct): The company-wide benchmark for a single fibernode's ability to attract and convert potential customers.
- Underperforming Fibernodes: Nodes identified by comparing their specific utilization rate against the national network average.

	fibernode character varying (200)	region character varying (200)	capex_cost numeric	util_pct numeric	network_avg_util_pct numeric	status text
1	F737809	Jakarta Barat	371021156	64.00	76.90	Underperforming
2	F419512	Jakarta Selatan	367678612	76.00	76.90	Underperforming
3	F889057	Jakarta Timur	363366481	76.00	76.90	Underperforming
4	F926002	Jakarta Utara	357473123	66.00	76.90	Underperforming

Based on the data, four regions in Jakarta (West, South, East, and North) are currently flagged as 'Underperforming'. This indicates that the subscriber penetration in these high-investment areas remains below the national benchmark, suggesting an opportunity for targeted growth or operational review.

3. Compare Monetization Trend Between Exclusive vs Open Access Areas and FTTH vs HFC

This section explain trend monetization between Exclusive vs Open Access Area and FTTH vs HFC as explained below:

	snapshot_date character varying (200)	technology character varying (200)	area_type text	gross_revenue bigint	contribution_margin bigint	margin_pct numeric
1	2025-12-31 0:00:00	FTTH	Exclusive	14282000	-1410000	-9.87
2	2025-12-31 0:00:00	HFC	Exclusive	5766000	-482000	-8.36
3	2025-12-31 0:00:00	FTTH	Open Access	57162000	-8318000	-14.55
4	2025-12-31 0:00:00	HFC	Open Access	25718000	-4078000	-15.86
5	2025-11-30 0:00:00	HFC	Exclusive	6092000	-482000	-7.91
6	2025-11-30 0:00:00	FTTH	Exclusive	14834000	-1538000	-10.37
7	2025-11-30 0:00:00	HFC	Open Access	26024000	-4122000	-15.84
8	2025-11-30 0:00:00	FTTH	Open Access	58012000	-8444000	-14.56
9	2025-10-31 0:00:00	FTTH	Exclusive	13100000	-1558000	-11.89
10	2025-10-31 0:00:00	HFC	Exclusive	4876000	-426000	-8.74
11	2025-10-31 0:00:00	FTTH	Open Access	55368000	-8428000	-15.22
12	2025-10-31 0:00:00	HFC	Open Access	26342000	-4164000	-15.81
13	2025-09-30 0:00:00	FTTH	Exclusive	12360000	-1602000	-12.96
14	2025-09-30 0:00:00	HFC	Exclusive	4744000	-532000	-11.21
15	2025-09-30 0:00:00	HFC	Open Access	25884000	-4248000	-16.41
16	2025-09-30 0:00:00	FTTH	Open Access	54970000	-8556000	-15.56
17	2025-08-31 0:00:00	HFC	Exclusive	4458000	-498000	-11.17
18	2025-08-31 0:00:00	FTTH	Exclusive	11256000	-1504000	-13.36
19	2025-08-31 0:00:00	FTTH	Open Access	54468000	-8684000	-15.94
20	2025-08-31 0:00:00	HFC	Open Access	26210000	-4272000	-16.30

As observed,Exclusive areas using FTTH dan HFC have highest revenue among all while on the other hand Open Access areas using FTTH technology show a trend of increasing gross revenue, yet they consistently yield a negative contribution margin (averaging -15%). This is driven by the fact that

the Total Cost (comprising ao_wholesale_price and lease_fee_per_active) exceeds the Gross Revenue (Package Price + ao_rrp_price). This phenomenon is likely caused by high Lease Fees and aggressive price wars among Service Companies (ISPs) in Open Access areas, forcing retail prices down to a point that does not yet cover the infrastructure and third-party content costs.

4. Structural Patterns in Insfrastructure Performance

From the findings above, we can conclude that:

- While HFC technology maintains a higher penetration rate, FTTH consistently generates significantly higher gross revenue across all regions. The company can strategically leverage HFC to maintain market share (customer retention) while utilizing FTTH as the primary driver for revenue growth.
- The combination of High CAPEX and Low Utilization in urban areas, such as West, South, East, and North Jakarta, indicates a highly saturated market. This suggests that customers have numerous options, leading to intense competition among providers.
- The market structure in Open Access areas forces the company into aggressive pricing wars. This results in gross revenues that fail to cover lease and packaging fees (negative margins). Achieving operational efficiency and cost optimization is critical for long-term sustainability in these areas.

Section 2 – Servco Performance & Competition

1. Servco Performance Analysis

Servco Performance Analysis evaluates the operational efficiency of each service company across several key metrics. To provide a clear ranking of market success, the following list is sorted in descending order based on the penetration rate.

This analysis focuses on three core pillars:

- Active CA Growth: Tracking the monthly increase in active subscribers to identify market momentum.
- Lease Revenue Contribution: Measuring the total financial contribution from each Servco based on their active user base.
- Market Penetration: Assessing the ratio of active customers against total homepasses to determine market dominance in specific regions.

The results are as follows:

	snapshot_date character varying (200)	servco_name character varying (200)	active_ca bigint	active_ca_growth bigint	lease_revenue_contribution bigint	penetration_rate numeric
1	2025-09-30 0:00:00	XLS	182	6	25480000	67.66
2	2025-10-31 0:00:00	XLS	179	-3	25060000	66.54
3	2025-11-30 0:00:00	XLS	178	-1	24920000	66.17
4	2025-08-31 0:00:00	XLS	176	8	24640000	65.43
5	2025-12-31 0:00:00	XLS	175	-3	24500000	65.06
6	2025-07-31 0:00:00	XLS	168	7	23520000	62.45
7	2025-11-30 0:00:00	MMA	165	7	21450000	61.80
8	2025-11-30 0:00:00	SURGE	160	15	20000000	61.54
9	2025-12-31 0:00:00	MMA	162	-3	21060000	60.67
10	2025-12-31 0:00:00	SURGE	157	-3	19625000	60.38

11	2025-06-30 0:00:00	XLS	161	13	22540000	59.85
12	2025-10-31 0:00:00	MMA	158	6	20540000	59.18
13	2025-11-30 0:00:00	TETRA	152	9	19456000	57.79
14	2025-09-30 0:00:00	MMA	152	0	19760000	56.93
15	2025-08-31 0:00:00	MMA	152	15	19760000	56.93
16	2025-12-31 0:00:00	TETRA	149	-3	19072000	56.65
17	2025-06-30 0:00:00	MMA	118	10	15340000	55.92
18	2025-10-31 0:00:00	SURGE	145	10	18125000	55.77
19	2025-05-31 0:00:00	XLS	148	22	20720000	55.02
20	2025-10-31 0:00:00	TETRA	143	6	18304000	54.37
21	2025-06-30 0:00:00	SURGE	105	14	13125000	53.30
22	2025-11-30 0:00:00	INFOCOM	128	12	17280000	53.11
23	2025-06-30 0:00:00	TETRA	107	5	13696000	52.97
24	2025-12-31 0:00:00	INFOCOM	126	-2	17010000	52.28
25	2025-09-30 0:00:00	TETRA	137	4	17536000	52.09

2. Servco 101 Performance and Penetration Against Minimum Guarantee

- Actual_penetration_pct is the ratio of homes with an ACTIVE status relative to the total available homepasses. This metric measures how effectively a Servco is capturing the market in its assigned area.
- Guarantee_target_pct is the minimum penetration threshold (25%) that a Servco is contractually obligated to achieve within its territory.
- The 'Achieved' status is granted when the actual penetration exceeds or meets the guarantee target, indicating that the Servco is performing according to or above expectations. Conversely, if the actual penetration falls below this target, it signals an underperformance that may require further strategic evaluation.

	snapshot_date character varying (200)	servco_name character varying (200)	actual_penetration_pct double precision	guarantee_target_pct double precision	status text
1	2025-01-31 0:00:00	XLS	32.342007434944236	25	Achieved
2	2025-02-28 0:00:00	XLS	36.43122676579926	25	Achieved
3	2025-03-31 0:00:00	XLS	40.520446096654275	25	Achieved
4	2025-04-30 0:00:00	XLS	46.84014869888476	25	Achieved
5	2025-05-31 0:00:00	XLS	55.01858736059479	25	Achieved
6	2025-06-30 0:00:00	XLS	59.85130111524164	25	Achieved
7	2025-07-31 0:00:00	XLS	62.45353159851301	25	Achieved
8	2025-08-31 0:00:00	XLS	65.4275092936803	25	Achieved
9	2025-09-30 0:00:00	XLS	67.65799256505576	25	Achieved
10	2025-10-31 0:00:00	XLS	66.54275092936803	25	Achieved
11	2025-11-30 0:00:00	XLS	66.17100371747212	25	Achieved
12	2025-12-31 0:00:00	XLS	65.05576208178438	25	Achieved

3. Performance Comparision In Exclusive Window vs Post Exclusive and In single-ISP vs multi-ISP areas

Exclusive Window is defined as a period where each Home ID is served by only one ISP, meaning customers have a single option for their fiber optic provider. Conversely, Post-Exclusive occurs when a Home ID has ple fiber optic company options, introducing market competition. The

Average Penetration Percentage represents the ratio of active subscribers to total homepasses in these areas. The results are as follows:

	exclusivity_status text	active_customers bigint	avg_penetration_pct numeric
1	Exclusive Window/Single ISP Area	712	7.76
2	Post-Exclusive/Multi ISP Area	182	8.97

Section 3 – Media Monetization & Profitability

1. Media attach rate per servco and per region

	region character varying (200)	servco_name character varying (200)	total_internet_subs bigint	total_media_subs bigint	attach_rate_pct numeric
1	Bekasi	INFOCOM	10	6	60.00
2	Bekasi	SURGE	30	16	53.33
3	Bekasi	XLS	22	11	50.00
4	Bekasi	MMA	26	13	50.00
5	Bekasi	TETRA	15	4	26.67
6	Depok	SURGE	17	10	58.82
7	Depok	INFOCOM	21	12	57.14
8	Depok	MMA	27	15	55.56
9	Depok	XLS	20	9	45.00
10	Depok	TETRA	13	3	23.08
11	Jakarta Barat	MMA	23	14	60.87
12	Jakarta Barat	TETRA	24	14	58.33
13	Jakarta Barat	XLS	32	14	43.75
14	Jakarta Barat	SURGE	22	9	40.91
15	Jakarta Barat	INFOCOM	21	6	28.57
16	Jakarta Pusat	MMA	13	9	69.23
17	Jakarta Pusat	SURGE	19	12	63.16
18	Jakarta Pusat	XLS	24	13	54.17
19	Jakarta Pusat	INFOCOM	9	3	33.33
20	Jakarta Pusat	TETRA	13	4	30.77
21	Jakarta Selatan	MMA	23	16	69.57
22	Jakarta Selatan	SURGE	27	16	59.26
23	Jakarta Selatan	INFOCOM	17	8	47.06
24	Jakarta Selatan	XLS	22	10	45.45
25	Jakarta Selatan	TETRA	28	11	39.29
26	Jakarta Timur	TETRA	34	17	50.00
27	Jakarta Timur	INFOCOM	27	13	48.15
28	Jakarta Timur	MMA	33	15	45.45
29	Jakarta Timur	XLS	27	12	44.44
30	Jakarta Timur	SURGE	13	5	38.46
31	Jakarta Utara	TETRA	26	12	46.15
32	Jakarta Utara	INFOCOM	24	11	45.83

33	Jakarta Utara	XLS	43	17	39.53
34	Jakarta Utara	SURGE	31	12	38.71
35	Jakarta Utara	MMA	17	6	35.29
36	Tangerang	XLS	17	11	64.71
37	Tangerang	INFOCOM	17	9	52.94
38	Tangerang	SURGE	20	10	50.00
39	Tangerang	MMA	27	12	44.44
40	Tangerang	TETRA	20	5	25.00

2. Contribution analysis by Lease Revenue vs Media Revenue and Media Margin by Product

A. Lease Revenue vs Media Revenue

Lease Revenue represents the company's income generated from infrastructure rental costs, while Media Revenue accounts for earnings from customer media purchases. The table is presented in descending order based on each Servco's total contribution relative to the company's overall revenue.

	servco_name character varying (200)	total_lease_revenue numeric	total_media_revenue bigint	total_combined_revenue numeric	contribution_pct numeric
1	XLS	250180000	117863000	368043000	26.30
2	MMA	196040000	115032000	311072000	22.23
3	TETRA	172672000	80697000	253369000	18.10
4	SURGE	161375000	87865000	249240000	17.81
5	INFOCOM	149580000	68193000	217773000	15.56

B. Media Margin by Product

Media margin was calculated based on CATV Package and Add On Products

	catv_package character varying (200)	total_product_revenue bigint	total_actual_margin bigint	margin_pct numeric
1	CATV Joy Pro	26516000	20761000	78.30
2	CATV Joy Premium	17151000	13576000	79.16
3	CATV Joy Value	17040000	13020000	76.41

Media Margin by CATV Package

	add_on_product character varying (200)	total_product_revenue bigint	total_actual_margin bigint	margin_pct numeric
1	News Pack	10798000	9358000	86.66
2	Viu	10570000	8570000	81.08
3	Sport Pack	12353000	9518000	77.05
4	Catchplay	15443000	11563000	74.88
5	HBO Max	11543000	8348000	72.32

Media Margin by Add On Products

3. Identification on Most Profitable Media Product, Servco with Strongest Upsell Capability, and Regions with Highest Monetization Potential

The Most Profitable Media Products identified are CATV Joy Premium with the News Pack add-on, yielding a margin of 86.66%, followed by Viu at 81.08%.Furthermore, the Strongest Upsell Capability is measured by the ratio of ACTIVE media subscribers relative to the total active subscriber base. The results are as follows:

	servco_name character varying (200)	total_internet_subs bigint	total_media_subs bigint	upsell_success_rate_pct numeric
1	MMA	189	100	52.91
2	SURGE	179	90	50.28
3	XLS	207	97	46.86
4	INFOCOM	146	68	46.58
5	TETRA	173	70	40.46

Regions with Highest Monetization Potential are:

	region character varying (200)	total_homepass bigint	exclusive_flag character varying (200)	current_attach_rate numeric
1	Jakarta Utara	100	N	38.61
2	Jakarta Timur	100	N	46.39
3	Tangerang	100	N	46.53
4	Jakarta Barat	100	N	47.19
5	Bekasi	100	N	48.54
6	Depok	100	N	50.00
7	Jakarta Selatan	100	N	52.75
8	Jakarta Pusat	50	Y	44.12
9	Jakarta Barat	50	Y	45.45
10	Jakarta Timur	50	Y	45.95
11	Jakarta Utara	50	Y	47.50
12	Jakarta Selatan	50	Y	50.00
13	Jakarta Pusat	50	N	59.09

Jakarta Utara and Jakarta Timur are identified as the regions with the highest monetization potential, totaling 100 homepasses each but with relatively low attach rates (38.61% and 46.39%, respectively).

4. Is media bundling meaningfully improving ARPU?

ARPU (Average Revenue Per User) represents the average income generated from each customer over a specific period. In this analysis, we evaluate whether bundling significantly improves ARPU compared to internet-only subscriptions. The result shows in a table as follow:

	customer_segment text	avg_arpu numeric
1	Bundled (Internet + Media)	271726
2	Internet Only	132017

Bundled customers are defined as those registered in both the subscription_snapshot and media_package tables, whereas Internet Only customers are those found exclusively in the subscription_snapshot table without any media add-ons. The results indicate that the average ARPU for bundled customers is notably higher, confirming that media bundling meaningfully drives ARPU growth.

Section 4 – Strategic Recommendations

1. 3 Strategies priorities for the next 6 months
   A. Maximize monetization potential by targeting Jakarta Utara (Non-Exclusive), which currently shows a significant 20% attach rate deficit compared to the top-performing Jakarta Pusat (59.09%). With 100 homepasses available, we will replicate Jakarta Pusat's successful sales model and deploy 'Aggressive Bundling Packages' to capture the massive untapped potential in North Jakarta's open market.
   B. Capitalize on the high-performing MMA and SURGE Servcos, which lead the portfolio with upsell success rates of 52.91% and 50.28%, respectively. Furthermore, we will conduct a comprehensive sales process audit on MMA to identify best practices and bridge the 12% performance gap observed in TETRA.
   C. Launch a 'Bridge the Gap' campaign specifically targeting 213 internet-only subscribers across XLS and TETRA. By offering a 3-month 'Premium Media Trial,' we aim to convert these users into bundled segments. This transition is highly lucrative, as the minimal cost of media integration is offset by a 2x ARPU gain.

2. Strategic Action
   Based on our findings, FiberCo should pursue aggressive FTTH expansion through strategic regional targeting. While Multi-ISP areas show higher penetration rates (8.97% vs 7.76%), Single ISP areas currently drive larger scale with 712 active customers. Furthermore, we must prioritize bundling packages, as they significantly outperform internet-only plans by doubling the Average ARPU from 132,017 to 271,726.

Tableau Link:
https://public.tableau.com/newWorkbook/42b519a7-81b9-40bb-8a28-3b0516077f85#1